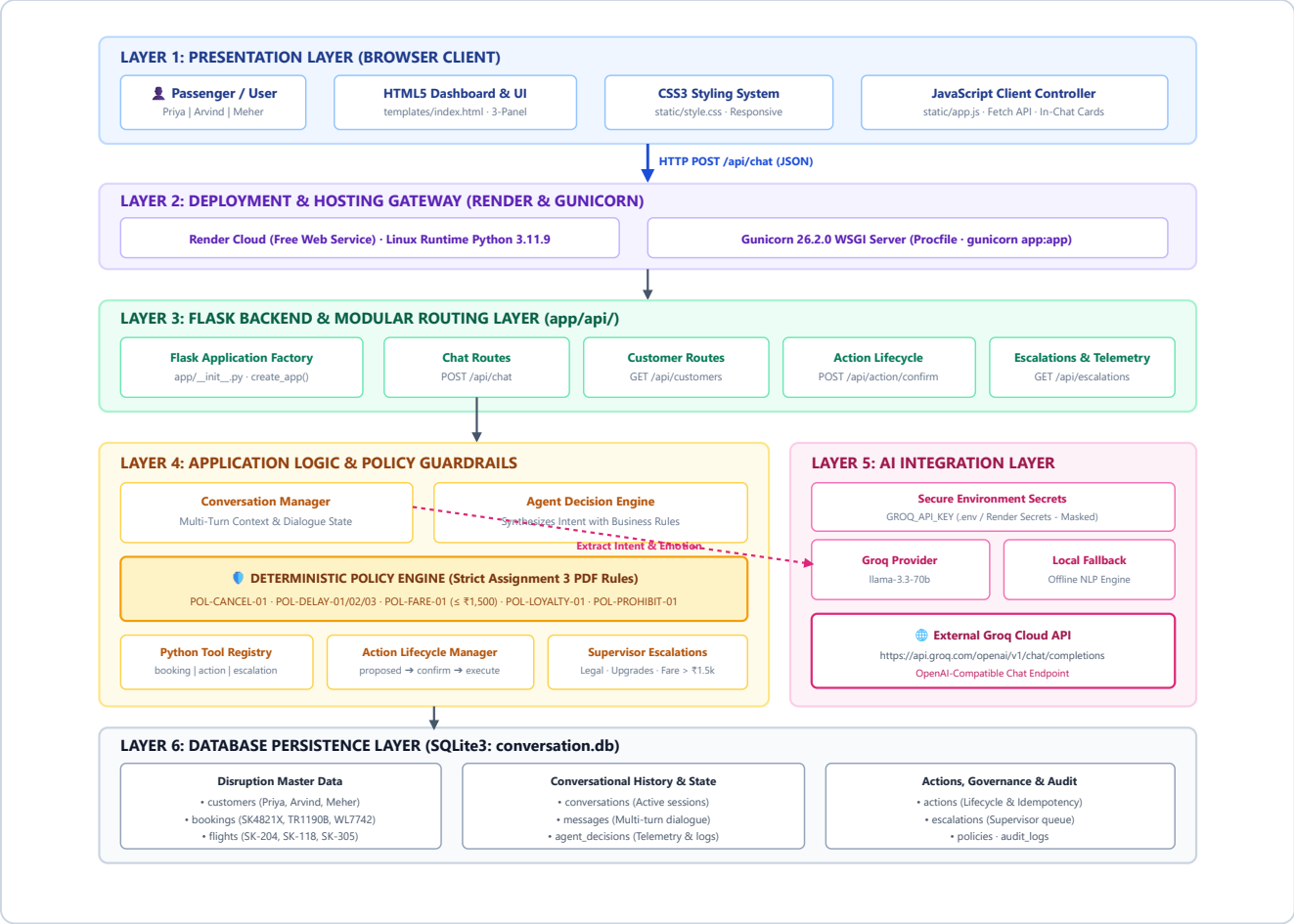


1. COMPLETE END-TO-END SYSTEM ARCHITECTURE DIAGRAM



2. TECHNICAL ARCHITECTURE & COMPONENT MAPPING

| Layer             | Component / File                                      | Implementation Details                                                                                     | Security & Governance                                                                    |
|-------------------|-------------------------------------------------------|------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------|
| 1. Presentation   | templates/index.html<br>static/style.css, app.js      | Responsive 3-panel UI, in-chat action confirmation cards, decision telemetry tab, scenario test runners.   | Zero API keys exposed in browser; all operations routed through backend.                 |
| 2. Deployment     | render.yaml, Procfile<br>runtime.txt (Python 3.11.9)  | Render Cloud Free Web Service running Gunicorn 26.2.0 WSGI server ('gunicorn app:app').                    | Ephemeral container isolation; auto-restarts and auto-seeds database cleanly.            |
| 3. API Routing    | app/api/chat_routes.py<br>customer, action, esc       | Flask 3.1.0 Application Factory with modular REST blueprints. Dispatches requests to logic services.       | Input validation, JSON schema enforcement, sanitization against arbitrary injections.    |
| 4. Logic & Safety | app/agents/, policies/<br>services/, tools/           | ConversationManager maintains context; DecisionEngine coordinates; PolicyEngine enforces strict PDF rules. | LLM cannot override policy. Strict two-stage confirmation for refunds and rebookings.    |
| 5. AI Integration | app/ai/groq_provider.py<br>local_fallback_provider.py | Groq Cloud REST API ('llama-3.3-70b-versatile' via '/v1/chat/completions') + zero-dependency local engine. | 'GROQ_API_KEY' stored exclusively in environment variables / Render secrets.             |
| 6. Database       | app/database/<br>conversation.db (SQLite)             | 10 relational tables: customers, bookings, flights, policies, messages, actions, escalations, audit.       | Parameterized SQL queries (prevents SQL injection); idempotency keys prevent duplicates. |

3. END-TO-END REQUEST-RESPONSE QUERY WORKFLOW

1 Passenger Input & Session Context

Customer selects profile (e.g., Priya Nair, Gold) and submits query: "My flight was cancelled. I want a refund and free upgrade to business class."

2 Client Dispatch (Fetch API)

static/app.js packages passenger ID, message, and session state into JSON and issues an HTTP POST request to /api/chat.

3 Web Gateway & Flask Factory

Render forwards traffic to Gunicorn, which dispatches to Flask. ConversationManager retrieves prior dialogue turns and disruption facts.

4 Natural Language Understanding (Groq)

Backend loads GROQ\_API\_KEY from env variables and sends REST request to Groq Cloud. Returns structured intent, entities, emotion, and urgency.

5 Policy Guardrail Evaluation

PolicyEngine enforces Assignment 3 rules: grants full refund eligibility (POL-CANCEL-01), denies cabin upgrade (POL-LOYALTY-01), and triggers escalation.

6 Safe Tool & Action Execution

ActionService stages refund in awaiting\_confirmation status with idempotency key. EscalationService logs ticket in SQLite.

7 Persistence & Audit Logging

Dialogue turns, decision metadata, telemetry, and audit events are recorded in SQLite (conversation.db).

8 Interactive Client Rendering

Browser renders grounded response with policy citations, supervisor escalation pill, and an interactive Action Card with [✓ Approve] & [X Cancel] buttons.

4. CORE SECURITY & SAFETY PRINCIPLES

Key Architectural Invariants

- Zero LLM Authorization:** The LLM is strictly used for language understanding. It has no authority to invent policies, waive fees above ₹1,500, or approve out-of-policy upgrades.
- Two-Stage Confirmation:** Irreversible actions (refunds, rebooking) cannot execute on initial inquiry; they require explicit passenger confirmation.
- Secret Isolation:** All API keys (GROQ\_API\_KEY) are stored in server environment variables and never leaked to frontend JavaScript or Git.